

Module 15) Advance Python Programming

10. Search and Match Functions

- Using `re.search()` and `re.match()` functions in Python's `re` module for pattern matching.

1. `re.match()`

- Tries to **match a pattern only at the beginning** of the string.
- If the pattern matches at the start, it returns a **match object**.
- If it doesn't match at the start, it returns **None**.

Key point:

- Strictly checks **from the start** of the string.

2. `re.search()`

- Searches the **entire string** for the first occurrence of the pattern.
- Returns a **match object** if found **anywhere** in the string.
- Returns **None** if no match is found.

- **Difference between search and match.**

Aspect	re.match()	re.search()
Purpose	Checks for a match only at the start of the string	Searches for a match anywhere in the string
Matching behavior	Must match from the beginning (position 0)	Can match at any position in the string
Return value	Returns a match object if pattern matches at start, else None	Returns a match object if pattern found anywhere, else None
Use case	When you want to verify if the string starts with a pattern	When you want to find the pattern anywhere in the string

- Use re.match() to check **start of string** only.
- Use re.search() to find pattern **anywhere** in the string.